Project Design Phase-I
Solution Architecture

Date	08 March 2026
Team ID	
Project Name	Visualizing Housing Market Trends using Tableau & Flask
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

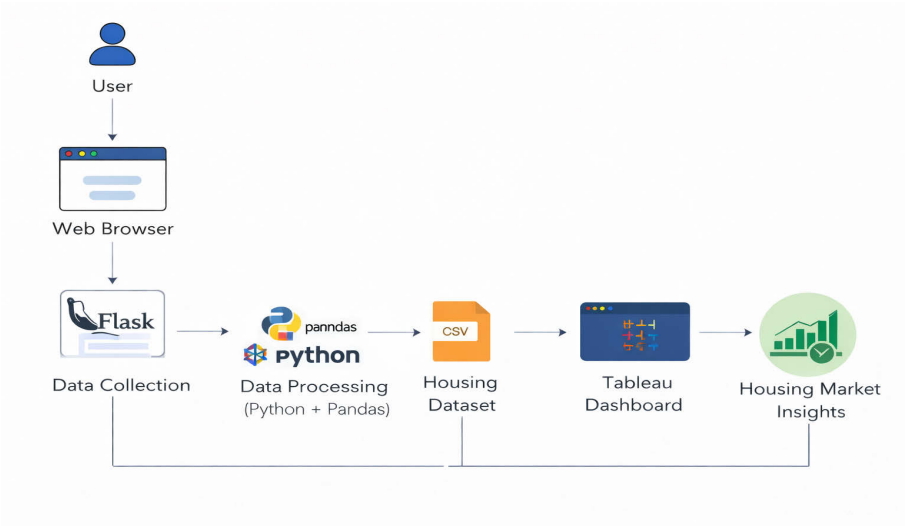

Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/

Solution Architecture

The proposed solution architecture focuses on analyzing housing market data and presenting insights through interactive visual dashboards. The system integrates data processing, visualization, and web technologies to provide an efficient platform for housing market analysis.

The architecture bridges the gap between raw housing datasets and meaningful business insights for real estate analysts and investors.

Goals of the Solution Architecture

• Identify the most effective technology stack to analyze housing market data.
• Provide an interactive visualization system to understand housing price trends.
• Define the structure and components of the application including data processing, visualization, and web interface.
• Deliver a scalable solution that can support future housing market analytics.

Solution Architecture Components

1 Data Source

Housing dataset containing attributes such as price, bedrooms, bathrooms, floors, renovation status, house age, and basement area.

2 Data Processing Layer

Python and Pandas are used to clean, transform, and prepare the housing dataset for analysis.

3 Visualization Layer

Tableau dashboards visualize housing price trends, house age distribution, renovation impact, and housing features.

4 Application Layer

A Flask web application embeds Tableau dashboards and provides users with an interface to interact with the visualizations.

5 User Layer

Real estate analysts, investors, and property managers access the dashboards through a web browser to explore housing market insights.